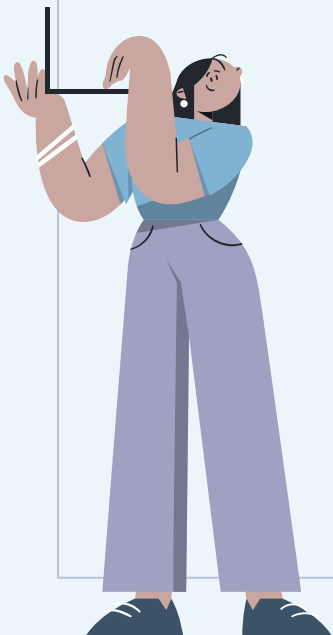


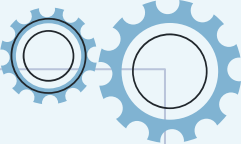
Time Series

ARIMA + GARCH

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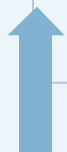
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ARMA + GARCH

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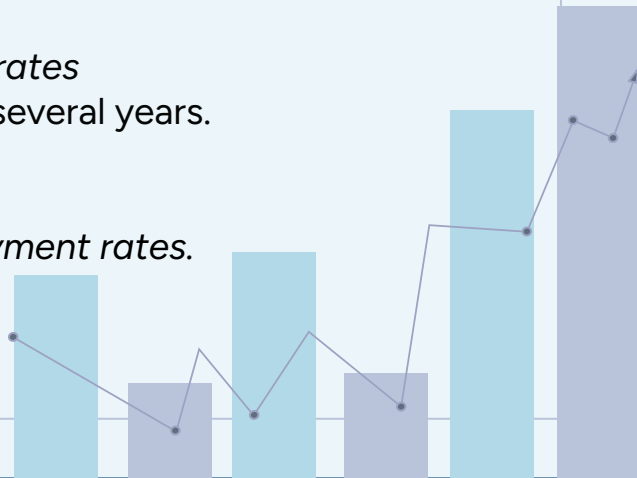


What is a Time Series?

- A set of data points collected, recorded, or measured in specific, chronological order over time (daily, monthly, yearly, etc.).
- Goal: identify patterns such as *trends*, *seasonality*, and *temporal dependence* so we can understand behavior and make forecasts about future values.

Real Life Applications:

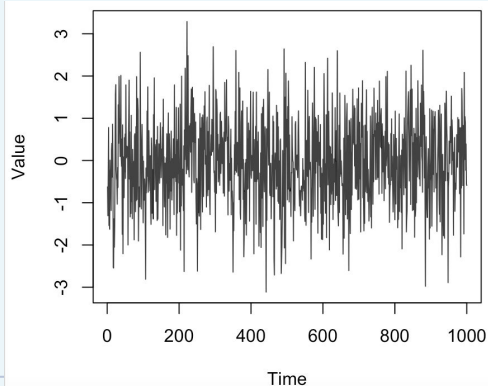
- **Finance:** model & forecast *stock prices, exchange rates, interest rates*
 - Daily stock prices (open, close, volume) of a company over several years.
- **Economics:** analyze *economic indicators, business cycles*.
 - Quarterly GDP growth, monthly inflation rates, or unemployment rates.
- **Public Health:** monitor and predict *disease spread, health trends*.
 - Weekly counts of influenza or COVID-19 cases over time.



Key Concepts

Stationarity

- Constant mean, constant variance, no systematic trend / changing seasonality.
- Example: White Gaussian Noise

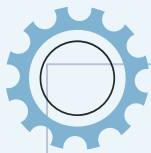


Autocorrelation

- Correlation between a time series and its past values (lags).
- Goal:
 - Detect dependence over time in the data
 - Identify patterns such as persistence or seasonality
 - Help select appropriate time series models (AR, MA, ARIMA)
- Example:

Stock returns have low autocorrelation, but volatility of returns often shows strong autocorrelation.



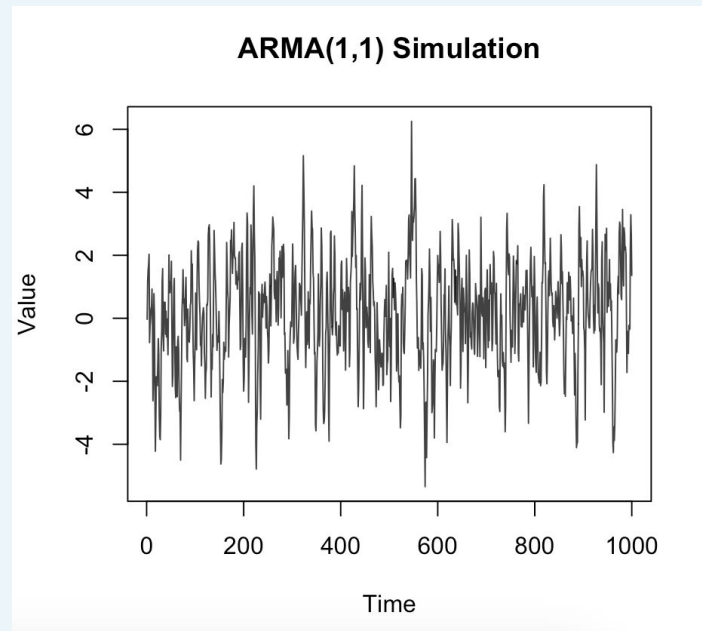


ARIMA Model

ARIMA was developed to:

- Model time dependence in data
- Handle trends and non-stationarity
- Improve forecasting accuracy

Purpose: Describe pattern in time series and predict future values





Correlated Errors: Regression vs Time Series

Model Type	Errors Assumption	What happens in Reality?	Example
Ordinary Regression	Independent errors	Works well when observations are unrelated	Predicting house prices using size, location, etc. One error does not affect the next observation
Time Series	Time-dependent observations	Errors often become correlated if time dependence is ignored	Daily product demand: if demand is high today, it is likely high tomorrow
ARIMA Approach	Model time dependence	Reduces correlated errors by using past values and past shocks	Monthly sales forecasting using past sales patterns



ARIMA(p,d,q)

ARIMA = AR + I + MA

$$y'_t = c + \phi_1 y'_{t-1} + \cdots + \phi_p y'_{t-p} + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q} + \epsilon_t,$$

AR (Auto Regressive)

- Current value depends on past values
- Parameter: **p** = AR order

$$X_t = \phi_1 X_{t-1} + \epsilon_t$$

I (Integration)

- Making the series stationary by differencing
- Parameter: **d** = number of difference
- 1st difference: $y'_t = y_t - y_{t-1}$
- 2nd difference:

$$\begin{aligned} y''_t &= y'_t - y'_{t-1} \\ &= (y_t - y_{t-1}) - (y_{t-1} - y_{t-2}) \\ &= y_t - 2y_{t-1} + y_{t-2}. \end{aligned}$$

MA (Moving Average)

- Current value depends on past errors.
- Parameter: **q** = MA order

$$X_t = \epsilon_t + \theta_1 \epsilon_{t-1}$$

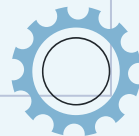


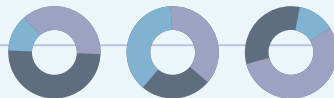
SARIMA(p,d,q)(P,D,Q)_s

$$\Phi_P(B^s) \phi_p(B) (1 - B)^d (1 - B^s)^D Y_t = \Theta_Q(B^s) \theta_q(B) \varepsilon_t$$

B	Backshift operator ($BY_t = Y_{t-1}$)
$(1 - B)^d$	Regular differencing (removes trend)
$(1 - B^s)^D$	Seasonal differencing
$\phi_p(B)$	Non-seasonal AR polynomial
$\theta_q(B)$	Non-seasonal MA polynomial
$\Phi_P(B^s)$	Seasonal AR polynomial
$\Theta_Q(B^s)$	Seasonal MA polynomial
ε_t	white noise error

- Many time series contain **seasonal patterns** that repeat regularly.
- Examples:
 - **Ice cream sales** increase every summer
 - **Electricity demand** rises every Monday
 - **Retail sales** spike every December
- SARIMA removes seasonal patterns using **seasonal differencing**.
Example:
If data is **monthly with yearly seasonality**: $Y_t - Y_{t-12}$
This compares values with the **same month last year**.





GARCH(p,q)

Generalized Autoregressive Conditional Heteroskedasticity

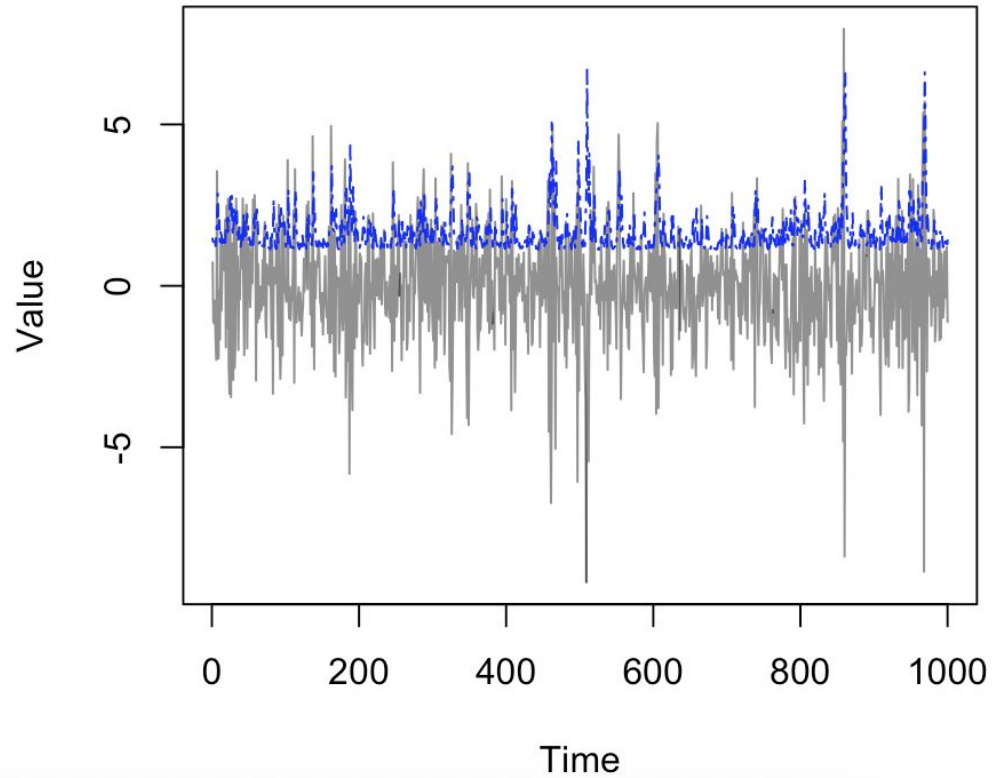
- *Examples:*
 - Dot-com bubble crash (2000)
 - Global financial crisis (2008)
- Model how variance changes over time (Heteroskedasticity)
- Predict future volatility with recent volatility

- GARCH Formula:
$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \epsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$$

Parameter:

- **p**= GARCH order (past volatility)
- **q**= ARCH order (past shocks)

GARCH(1,1) Simulation





Mean Reversion in GARCH

In GARCH (1,1) model, volatility evolve as:

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

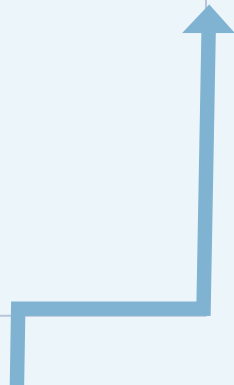
This means current volatility depends on:

recent shocks (ϵ_{t-1}^2) and **previous volatility** (σ_{t-1}^2)

Condition for Mean Reversion: $\alpha + \beta < 1$

Long Run Variance:

$$\frac{\omega}{1 - \alpha - \beta}$$



ARMA + GARCH

- ARMA: models average behavior
GARCH: models volatility (changing risk)
- In financial time series, often shows:
 - Returns : weak autocorrelation
 - Squared returns: strong autocorrelation (volatility clustering)
- Use GARCH alone, when:
 - Mean is constant
 - Return shows little autocorrelation (uncorrelated but volatility clusters)
- Use **ARMA + GARCH** when:
 - The mean of the series **shows autocorrelation**
 - The variance changes over time (**heteroscedasticity**)
- Steps:
 1. Fit **ARMA** to model the mean.
 2. Check **residuals**.
 3. If squared residuals show autocorrelation -> apply **GARCH**.



Thanks!

